Lesson notes · for you

Fair or unfair models

Data science · Level 4 · Lesson 3

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Find bias with error rates by group.
- Explain where bias starts.
- Plan checks before adopting a model.

Activity: Fair for everyone?

1. Warm-up (5 min): Say whether an app that is right 90% of the time must be fair to everyone.
2. Count (10 min): With a partner, work out each group's error rate: town pupils 3 wrong in 50, village pupils 11 wrong in 50.
3. Fix (10 min): List what could cause the gap and two ways to make the model fairer.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Which group had more errors?
- Where does bias come from?
- Who should check a model's results?

Check yourself

- Group A: 3 wrong in 20. Group B: 9 wrong in 20. Write both error rates.

Try it this week
Try a voice assistant or a photo app with different voices or faces in your family. Does it work equally well for everyone?

Worksheet

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Name _____ Class _____ Date _____

1 **Where does most bias in a model start?**

◯ In the data it learned from ◯ In the pupils ◯ In the screen colour

2 **A model can still discriminate by gender after the gender column is deleted.**

◯ True ◯ False

3 **Finish fnr(r, p) so it returns the false negative rate as a whole percentage, then print it for both groups.**

```
def fnr(r, p):
    return 0

ga = fnr([1, 1, 1, 1], [1, 1, 1, 0])
gb = fnr([1, 1, 1, 1], [1, 0, 0, 0])
print(ga, gb)
```

Write your program:

4 **Match each source of bias to an example.**

Under-representation	• • • •	Past hiring decisions used as “good candidate”
Biased labels	• • • •	Postcode standing in for income
Proxy	• • • •	Few rural voices in training data

5 **Base rates differ between two groups. What is generally true?**

◯ You cannot equalise error rates and calibration at once ◯ Accuracy settles the question

◯ Any model is automatically fair

✎ **Has an app ever not understood you, or someone you know? Who do you think it learned from?**

Check your answers

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1 **In the data it learned from**

A model is only as fair as its training examples.

2 **True**

Other features can act as proxies for it.

3

```
def fnr(r, p):
    pos = [b for a, b in zip(r, p)
           if a == 1]
    miss = pos.count(0)
    pct = 100 * miss / len(pos)
    return round(pct)

ga = fnr([1, 1, 1, 1], [1, 1, 1, 0])
gb = fnr([1, 1, 1, 1], [1, 0, 0, 0])
print(ga, gb)
```

Group A misses 1 positive in 4 (25%), group B misses 3 in 4 (75%).

4 **Under-representation — Few rural voices in training data**
Biased labels — Past hiring decisions used as "good candidate"
Proxy — Postcode standing in for income

Each injects a group difference into the model.

5 **You cannot equalise error rates and calibration at once**

Fairness criteria conflict, so people must choose deliberately.